This CVPR paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

EfficientVPR: Toward Efficient Visual Place Recognition via Scene-Aware Prompt Tuning and Adaptive Feature Enhancement

Wenjing Tang[1,2] Chuanguang Yang[1*] Zhulin An[1*] Libo Huang[1] Yongjun Xu[1]
[1]State Key Laboratory of AI Safety, Institute of Computing Technology, Chinese Academy of Sciences
[2]University of Chinese Academy of Sciences, Beijing, China
{tangwenjing25@mails.ucas.ac.cn}
{yangchuanguang, anzhulin, huanglibo, xyj}@ict.ac.cn

Abstract

Visual place recognition (VPR) faces critical challenges in handling extreme environmental variations while meeting the computational constraints of practical applications. Current methods predominantly address these challenges by either scaling up model capacity or employing computationally intensive reranking stages, creating a significant efficiency bottleneck. To overcome this limitation, we propose EfficientVPR, a lightweight one-stage framework that achieves speed-accuracy trade-offs through two key innovations: i) a scene-aware visual prompt tuning method which adapts pretrained features with less parameters while dynamically adjusting to sample-specific characteristics, and ii) an instance-dependent key local feature enhancement module that further reinforces discriminative regions. Comprehensive evaluations on Pitts250k, MSLS, Eynsham, AmsterTime and SVOX demonstrate that our method establishes a new SOTA for DINOv2-small models by outperforming all same-scale competitors, and delivers a 73× speedup with 60% lower-dimensional features while maintaining competitive (within 2.5% average R@1 gap) against the SOTA DINOv2-large-based two-stage method. Our code will be available at https://github.com/WiniTang/EfficientVPR.

1. Introduction

Visual Place Recognition serves as a fundamental technology for autonomous driving and robotic navigation [6, 54], localizing queries by matching their features against a prebuilt database. The core dilemma lies in balancing three competing demands: robustness to environmental changes, memory efficiency, and real-time inference speeds [35].

To address the first issue, a commonly adopted technique [37, 45, 53] is re-ranking, which requires storing key local features for geometric consistency verification (e.g. RANSAC [17]). However, this enhancement comes at the cost of increased memory requirements and significantly longer retrieval times. Consequently, one-stage methods have become an important research focus.

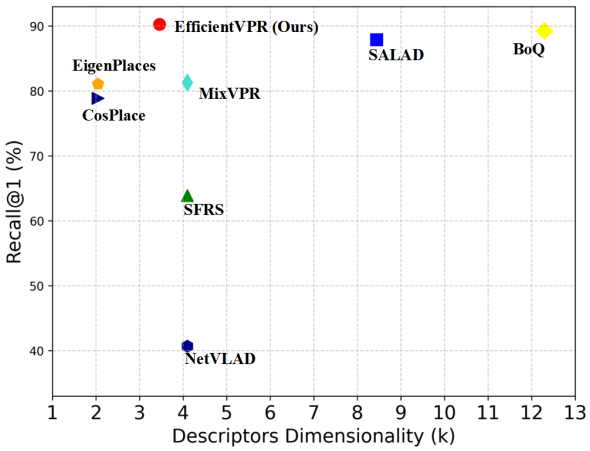


Figure 1. **Comparison with SOTA methods**: average R@1 over 7 datasets and descriptor dimensionality. Our method achieves the best performance with lower feature dimensions compared to approaches with similarly-scale.

These one-stage methods have therefore evolved to tackle environmental variations through three key strategies: backbone architecture redesign [9, 29, 30], model adaptation [4, 33], and advanced feature aggregation [2, 25, 39]. However, these methods still exhibit some limitations: i) They prioritize generic VPR features while overlooking sample-specific discriminative regions - the very features exploited by re-ranking methods. For instance, street or streetlight features can be either a background noise or a critical cue, depending on the context. ii) Existing methods struggle to adapt foundation models without either underutilizing their capacity or compromising their pre-trained features. iii) They tend to produce higher dimensionality of features, which increase the memory footprint dramatically.

* Corresponding author.

Hence, we propose a novel efficient one-stage VPR framework, which:

- Using compact yet powerful DINOv2-small [38] backbone instead of larger models to boost inference efficiency fundamentally.
- Introducing instance-dependent key local feature enhancement module that dynamically selects and enhances sample-specific features using backbone semantic priors to complement the backbone and enhance the model's discriminative power.
- Using designed SceneVPT with a small set of learnable prompts and an adaptive selection mechanism instead of adapter-based fine-tuning to achieve context-aware feature guidance with minimal computational overhead.

In summary, the key contributions of this paper include:

- We present a novel scene-aware visual prompt tuning method that achieves efficient adaptation of the entire pretrained model by introducing only minimal trainable parameters, while enabling sample-wise adaptive model through our adaptive prompt selection mechanism.
- We propose an instance-dependent key local feature enhancement module, which reduces one-stage models' sensitivity to non-overlapping regions during image encoding by emphasizing task-critical and sample-adaptive local features, even with limited model capacity.
- Comprehensive experimental results demonstrate that the proposed method achieves performance comparable to or better than state-of-the-art approaches while employing significantly more compact models, delivering faster retrieval speeds , and operating with lower-dimensional features, making it better suited for practical applications.

2. Related Work

2.1. Visual Place Recognition

Studies on VPR has undergone several methodological shifts, each striving to balance robustness, efficiency, and adaptability. Pioneering VPR methods [3, 8, 12, 28, 36, 43] rely on handcrafted global descriptors that aggregate low-level visual patterns using predefined statistical distributions, which gave way to deep learning-based methods that optimize feature extraction end-to-end, significantly improving discriminative power. Contemporary work can be broadly categorized into two main streams: one focuses on backbone redesign [29, 30, 33, 34], which encompasses the continuous refinement of backbone architectures and the adoption of different fine-tuning strategies, while another explores better way to utilize backbone-derived features [2, 4, 20, 25]. Despite these advances, a persistent limitation emerges across nearly all existing methods: the inability to dynamically adapt feature extraction to both task requirements and specific sample characteristics without incurring substantial computational overhead. Stems from these observations, our method achieves concurrent discriminative enhancement and model simplification through dynamic fusion of local and global features conditioned on instance-specific prompts processed by the backbone.

2.2. Parameter-Efficient Fine-Tuning in VPR

With large-scale pretrained models widely used, efficiently adapting them to downstream tasks attracts more attention [15, 24, 32, 46, 50, 51]. Parameter-efficient fine-tuning (PEFT) [19] has emerged as an effective solution. It aims to adapt model behavior by optimizing only minimal additional parameters. Some recent works [33, 34] in VPR has explored adapter-based methods [21, 22] to address the adaptation challenges of pretrained models. While these methods demonstrate improved efficiency over full fine-tuning, they exhibit fundamental limitations that prevent them from being ideal solutions, particularly for lightweight backbone adaptation. That is because these kind of method modify the original feature representations through direct alteration of entity feature tokens, which causes significant disruption to the pretrained knowledge. Visual prompt tuning [31] offers a lighter solution by adding learnable tokens that adjust features via Transformer attention. Due to the drastic and unpredictable visual appearance changes of a place, a single, static prompt [26] is forced to represent an "average" of all possible conditions, leading to a sub-optimal representation bias for specific query. Consequently, traditional prompt tuning is inadequate, necessitating a paradigm shift towards dynamic adaptation based on the unique characteristics of each input. To address this, we propose a novel scene-aware visual prompt tuning method (SceneVPT) that dynamically adapts prompts to both task objectives and environmental contexts. Some recent works [48, 49, 56] also recognize the importance of instance-aware prompt tuning, but most of them rely on external adapter to generate input related prompts. In contrast, our approach is simpler yet effective, requiring no extra modules.

3. Methodology

Fig. 2 illustrates the overall structure of our proposed method EfficientVPR, where scene-aware visual prompt tuning method SceneVPT and instance-dependent key local feature enhancement module are two core technical contributions that will be detailed in the following subsections.

3.1. Scene-aware Visual Prompt Tuning

To better leverage the potential of pretrained models and address pretrained models' semantic misalignment with downstream tasks, current methods like CricaVPR and SelaVPR employ adapter-based parameter-efficient fine-tuning method to fine-tune the backbone. While more efficient than full fine-tuning, their layer-wise module insertion

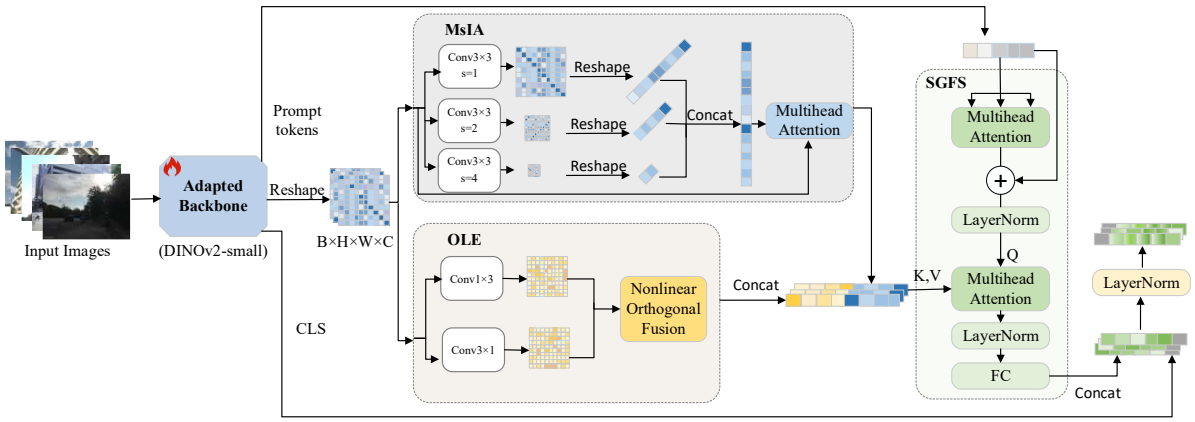


Figure 2. **The overall architecture of our one-stage method**. To maintain strong discriminative power while ensuring lightweight efficiency, we employ DINOv2-small as the backbone with SceneVPT for fine-tuning, preserving generalizable features while adapting to instance-specific characteristics. During feature enhancement, EfficientVPR uses MsIA and OLE to extract multi-scale global features and orientation-aware local features respectively, while SGFS performs instance-specific local feature enhancement of key regions based on sample-dependent semantic cues obtained from backbone.

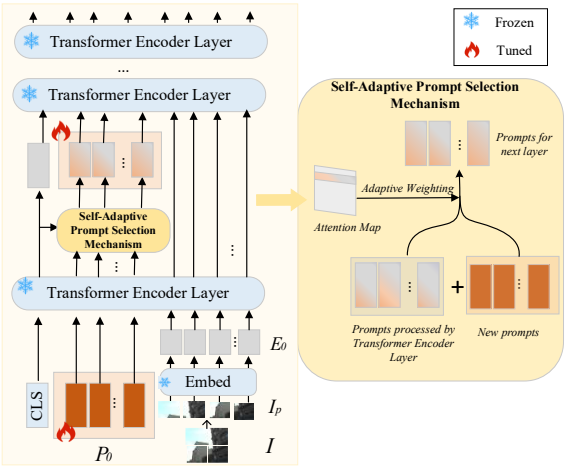


Figure 3. **SceneVPT Overview**. Our SceneVPT extends VPT-deep via adaptive prompt selection, dynamically adjusting to input characteristics based solely on the backbone's CLS token.

inevitably increases model complexity. More fundamentally, the adapter mechanism's inherent operation of directly modifying the entity feature tokens causes significant degradation to the pretrained feature space, consequently impairing the inherent generalizability of the pretrained backbone.

Inspired by VPT, we design our scene-aware visual prompt tuning method (SceneVPT), illustrated in Fig. 3. Unlike the original VPT, SceneVPT dynamically preserves sample-relevant learnable prompts from preceding layers while replacing irrelevant ones with new learnable prompts during layer-wise optimization of the pretrained model. This modification is motivated by following rationale:

- While VPT-deep introduces learnable prompts across layers and offers more flexible feature adaptation than its shallow counterpart, it remains largely static and scene-agnostic in practice. This limits its capacity to capture the unique and varying characteristics of individual input scenes, reinforcing the need for dynamic adaptation.
- The compact feature space of DINOv2-small heightens its susceptibility to catastrophic forgetting during fine-tuning. As shown by Aghajanyan et al. [1], fine-tuning operates in a low intrinsic dimensional subspace, which largely overlaps with the native representation space in already compact models. Dynamically adjusting the insertion degree of new prompt across layers enables graded fine-tuning, which not only adapts the model to visual place recognition tasks but also maximally preserves the original knowledge of the pretrained model, effectively mitigating catastrophic forgetting during fine-tuning.

Since in ViT [16] CLS token summarizes the global information of the entire image. In order to obtain the prompts that are valid for the current sample in each layer and avoid the introduction of additional parameters, we use the attention weights from the CLS token to the prompt—computed by the existing attention modules at each layer—to dynamically re-weight the prompts in the current layer. Specifically, for each layer, we successively insert N_p learnable prompts in front of the original input. For the prompt at the i-th layer, its filtering weight is denoted as $\boldsymbol{\alpha}_i = \left[\boldsymbol{\alpha}_{i1}, \boldsymbol{\alpha}_{i2}, \cdots, \boldsymbol{\alpha}_{iN_p}\right]$ and is computed by:

$$\boldsymbol{\alpha}_{ij} = \text{sigmoid}(\text{s}'_\text{j} - \gamma), \tag{1}$$

$$s'_j = s_j / \sum_{k=1}^{N_p} s_k, \tag{2}$$

with s_j being layer $(i-1)$'s averaged CLS-prompt attention weights, and γ the learnable threshold.

The final prompts $\widehat{P}_{i-1}$ used for fine-tuning, consists of two parts: the sample-relevant prompt Z^P_{i-1} and the newly introduced prompt P_i. Thus, the fine-tuning process with

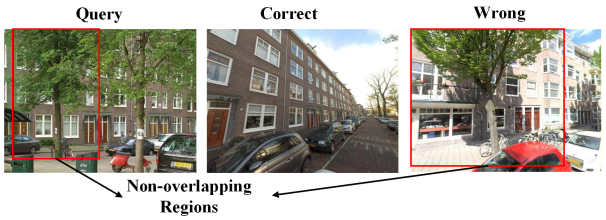


Figure 4. **Visualization of viewpoint-induced matching failure.** The first two images depict the same location, while the third shows a different but similar place. Dramatic viewpoint shifts and seasonal appearance variations cause trees in the query image (red border in the first picture) to become non-overlapping regions absent in the correct reference match. Current one-stage methods tend to be misled by these irrelevant features, producing false matches to incorrect locations with visually similar non-overlapping elements (red border in the third picture). By enhancing the discriminative representation of task-relevant and sample-specific key local regions and structural information, our method effectively mitigates this issue.

SceneVPT can be expressed by the following formula:

$$\left[Z_1^{CLS}, Z_1^P, Z_1\right] = L_1(\left[CLS, P_1, E_0\right]), \tag{3}$$

$$\left[Z_i^{CLS}, Z_i^P, Z_i\right] = \mathrm{L_i}(\left[Z_{i-1}^{CLS}, \widehat{P}_{i-1}, Z_{i-1}\right]), \mathrm{i} \geq 2, \tag{4}$$

$$\widehat{P}_{i-1} = \boldsymbol{\alpha}_i \cdot Z_{i-1}^P + \left(\mathbf{1}_{N_p} - \boldsymbol{\alpha}_i\right) \cdot P_i, i \geq 2. \tag{5}$$

where Z_i^{CLS} and Z_i denote the CLS token and the full output sequence from the i-th encoder layer L_i, respectively. E_0 is the patch sequence of the input image, Notably, N_p is the only hyperparameter that requires manual tuning.

3.2. Instance-dependent Key Local Feature Enhancement Module

Unlike two-stage methods, single-stage approaches primarily focus on capturing global structural information while neglecting critical local details. This limitation makes them particularly susceptible to interference from non-overlapping regions when encoding entire images in partially overlapping matching scenarios, ultimately leading to suboptimal performance under significant viewpoint variations, as illustrated in Fig. 4. This issue becomes more pronounced when reducing backbone capacity, since the constrained representational capacity further amplifies the model's reliance on dominant global patterns while struggling to preserve critical local features. Therefore, how to enhance discriminative representation of crucial local features so as to mitigate the sensitivity of global aggregation without employing reranking techniques emerges as the pivotal challenge for improving model efficiency.

To alleviate this issue, we design three sub-modules, which respectively handle multi-scale global features and local features, as well as compress and fuse the features extracted from both. We now detail these components.

3.2.1. Multi-scale Interaction Attention (MsIA)

For the feature sequence f_0 processed by backbone, it is first restored to the feature map f through spatial reconstruction. After that, we use a standard multi-scale structure to obtain features of different scales, and these feature maps are flattened and concatenated to form the multi-scale feature sequence $f_{multi} \in \mathbb{R}^{N' \times C}$, where $N' = \sum_{s \in [1,2,4]} \frac{H}{s} \times \frac{W}{s}$. Finally, we compute spatially adaptive weights via multi-head attention that dynamically fuses multi-scale contextual information by using f_0 as queries and the multi-scale feature sequence f_{multi} as keys and values.

3.2.2. Orientation-Aware Local Enhancer (OLE)

Considering that even in oblique views, local patterns and structural relationships remain preserved. It is necessary to further enhance these features. First of all, we reshape the feature sequence f_0 processed by backbone into feature map f in the same way as MsIA. The features are then processed to extract complementary structural cues through a dual-pathway design. Specifically, asymmetric convolutional kernels-a horizontal kernel (size 1×3) and a vertical kernel (size 3×1)-are used to captures local directional continuities that may be underrepresented in the global feature representation, yielding features f_h and f_v. To prevent simple aggregation of directional information, a nonlinear orthogonal fusion mechanism is applied. This mechanism first condenses the 2D feature map f_v into channel-wise scalar values f_v' via GeM pooling[40] and an MLP, producing a non-linear summary of vertical spatial statistics. It then computes the spatial correlation S between this descriptor and the horizontal features f_h through matrix multiplication, followed by an orthogonal projection decomposition to derive geometrically complementary features.

The complete process can be formulated as:

$$f_h' = f_h - (S/(\|f_v'\|_2 + \varepsilon)) \cdot f_v', \tag{6}$$

where $\|\cdot\|_2$ denotes L2 norm, ϵ is used to stabilize the value. Finally, we fuse orthogonalized features f_h' and f_v through 1×1 convolution and flatten the resulting feature map as the final local enhanced feature f_{OLE}.

3.2.3. Semantic-Guided Feature Selector (SGFS)

In order to achieve dynamic enhancement of the expression of key local features based on sample characteristics without using the costly local feature-based re-ranking technique, we utilize the sample-specific prompt processed by backbone as the semantic query signal and guide the feature screening process through the cross-attention mechanism.

Specifically, we denote the prompts processed by the final backbone layer as Z_L^P. Inspired by BoQ [4], we first adaptively refine the semantics of Z_L^P through self-attention mechanism, making it more focused on the key attributes

related to the task. This process can be expressed as:

$$f'_g = Z_L^P + \text{Attention}(\text{Z}_\text{L}^\text{P}, \text{Z}_\text{L}^\text{P}, \text{Z}_\text{L}^\text{P}), \tag{7}$$

$$f'_g = \text{LayerNorm}(\text{f}_\text{g}), \tag{8}$$

Then, for the concatenation feature $f_l \in \mathbb{R}^{2N \times C}$ of f_{MsIA} and f_{OLE}, we take f'_g as the query to perform cross-attention for enhancing task and sample specific features, and perform layer normalization on the results to stabilize the feature distribution. Formally, this can be expressed as:

$$f'_l = \text{Attention}(\text{f}'_\text{g}, \text{f}_\text{l}, \text{f}_\text{l}), \tag{9}$$

$$f_{out} = \text{LayerNorm}(\text{f}'_\text{l}). \tag{10}$$

Finally, to achieve more compact feature representations, we further compress f_{out} through a linear projection layer, reducing its feature dimension from N_P to 8.

4. Experiments

4.1. Datasets and Performance Evaluation

We conducted comprehensive evaluations on multiple mainstream benchmarks covering diverse scenarios, viewpoints, and conditional variations: Pitts250k [41] (substantial viewpoint variations), MSLS [47] (multi-scenario and multi-condition) , AmsterTime [52] (severe domain shift), Eynsham [14] (grayscale suburban images) and SVOX [11] (diverse weather and extreme illumination).

We adopt standard evaluation metrics using Recall@N - the percentage of queries with at least one correct reference image within the top-k results. A 25-meter threshold is applied for most datasets, while AmsterTime uses single-ground-truth evaluation, consistent with VPR benchmarks.

4.2. Implementation Details

Experiments were conducted on a single NVIDIA A800 GPU using PyTorch. Following GSV-Cities [5] training protocols, we trained our model on the GSV-Cities dataset with MS Loss [44] ($alpha = 1$, $beta = 50$, $lamda = 0$). The optimization employed a grouped strategy using Adam optimizer, with initial learning rates of $2e-4$ for main parameters and $2e-2$ for the pretrained model's fine-tuning parameters (decayed by $0.8\times$ every 5 epochs), for 30 epochs maximum. Training batches comprised 160 places with 4 randomly sampled 266×266 images per place. Test images were resized to 322×322. Our method achieves better performance despite using lower-resolution training inputs than outstanding models, which employ higher resolutions such as 480×640 (NetVLAD, CosPlace, EigenPlaces), 280×280 (BoQ) and 322×322 (FoL).

4.3. Comparisons with State-Of-The-Art Methods

We compare our model with a wide range of SOTA methods, including eight one-stage methods: NetVLAD, SFRS, CosPlace, MixVPR, EigenPlaces, CricaVPR, SALAD, BoQ and three two-stage methods: R2Former, SelaVPR and FoL. Tab.1 summarizes the results.

Despite using only a lightweight DINOv2-S backbone and a compact 3456-D descriptor, our model outperforms both: (1) all one-stage and two-stage methods at the same scale, and (2) even some methods employing larger-scale backbones. Specifically, with only 28% and 41% of the feature dimensions, our method achieves a 1.0% and 2.4% higher average R@1 than the one-stage SOTA BoQ and runner-up SALAD, respectively. Furthermore, on challenging datasets with severe domain and scene changes, our method surpasses BoQ by significant margins of 5.0% and 0.9% in R@1 on AmsterTime and Eynsham, respectively. When compared to the leading two-stage counterpart FoL, we achieve a 1.4% higher average R@1 with only 0.6% of its dimensions, notably outperforming it by 2.0% on MSLS-val (dramatic viewpoint changes) and 4.5% on SVOX-night (extreme illumination changes).

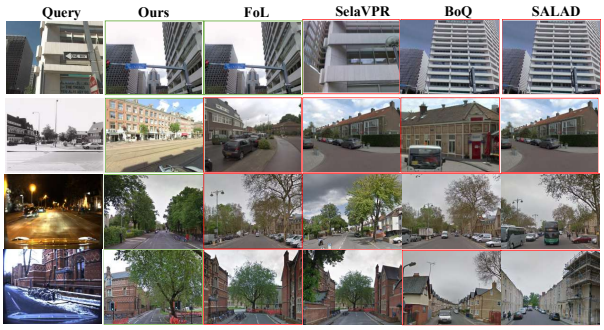


Figure 5. **Qualitative comparison on challenging cases**. Green boxes: correct matches; red boxes: errors. Drastic viewpoint changes along with domain shift (1th row), renovation-induced visual changes (2nd row), extreme lighting variations (3rd row), and severe occlusion coupled with time shift (4th row).

Comprehensive comparison across model scales. We further provide a comprehensive analysis by comparing the most competitive models from Tab.1 across backbone scales on key metrics, including average R@1, inference latency, model size, peak GPU memory, and training time. As shown in Tab. 2, our EfficientVPR establishes a compelling trade-off between efficiency and accuracy. Compared to SOTA one-stage and two-stage methods of similar scale, our EfficientVPR achieves the highest Avg. Acc., outperforming the top-performing BoQ by 1% with the fewest training parameters (7.7M) and the second-lowest latency (only slightly higher than the fastest method, SALAD, by 0.7ms, while $34.2\times$ faster than the two-stage method FoL). Although our method uses slightly more peak GPU memory than BoQ and SALAD with same scale, it significantly re-

Method	Backbone	Dim (k)	Pitts250k -test		MSLS -val		Amster Time		Eynsham		SVOX Night		SVOX Overcast		SVOX Snow	
			R@1	R@5	R@1	R@5	R@1	R@5	R@1	R@5	R@1	R@5	R@1	R@5	R@1	R@5
NetVLAD [7]	VGG-16	4	90.5	96.2	53.1	66.5	16.3	-	77.7	-	8.0	-	66.4	-	54.4	-
SFRS [18]	VGG-16	4	90.7	96.4	69.2	80.3	29.7	48.5	72.3	-	28.6	40.6	81.1	-	76.0	-
CosPlace [9]	ResNet-50	2	92.3	97.2	87.4	93.8	47.7	-	90.0	-	50.7	-	92.2	-	92.0	-
MixVPR [2]	ResNet-50	4	94.6	98.3	88.2	93.1	40.8	58.9	89.6	93.2	62.9	79.8	96.2	98.2	97.0	98.3
EigenPlaces [10]	ResNet-50	2	94.1	98.0	89.1	93.8	48.9	69.5	90.7	94.4	58.9	76.9	93.1	-	93.1	-
CricaVPR-16 [33] ‡	DINOv2-B	4	95.6	98.9	90.0	95.4	64.7	82.8	91.2	95.2	85.1	95.0	96.7	99.0	96.0	97.6
CricaVPR-1 [33] ‡	DINOv2-B	4	94.4	98.4	87.6	95.3	49.4	70.3	89.7	94.0	76.8	88.0	-	-	-	-
SALAD [25] †	DINOv2-S	8	94.0	98.4	90.0	96.0	53.9	76.0	90.7	94.4	90.0	96.7	97.9	98.7	98.6	99.2
BoQ [4] †	DINOv2-S	16	95.0	98.6	92.0	96.2	55.2	76.4	91.1	94.8	**94.7**	**98.4**	**98.4**	**99.4**	98.9	99.1
R2Former [55]	ViT-S	70	93.2	97.5	89.7	95.0	52.2	-	-	-	-	-	-	-	-	-
SelaVPR [34]	DINOv2-L	500	**95.7**	**98.8**	90.8	96.4	55.6	74.2	90.6	95.2	90.3	95.5	97.5	99.0	97.2	98.4
FoL [42] †	DINOv2-S	469	95.4	98.7	90.8	95.5	59.1	78.0	91.8	95.1	89.9	97.1	97.7	99.4	97.9	99.4
EfficientVPR (Ours)	DINOv2-S	3	95.4	**98.8**	**92.8**	**96.6**	**60.2**	**78.4**	**92.0**	**95.3**	94.4	97.9	**98.4**	**99.4**	**99.1**	**99.7**

Table 1. **Comparison with SOTA methods on seven benchmarks.** The best results are highlighted in **bold** and the second best are underlined. † Since the paper of SALAD, BoQ and FoL did not report their results on DINOv2-S, we additionally reproduced these results. ‡ CricaVPR utilizes a cross-image encoder architecture that exhibits instability with varying inference batch sizes. Consequently, we additionally provides results under the single query image scenario.

Method	Backbone	Dim	Train Time (min)	Latency per Query (ms)			Parameters(M)		Memory (GB)	Peak GPU Memory (MiB)	Avg. Acc.
				Extrac.	Rerank.	Total	Trainable	Total			
FoL †	DINOv2-L	5×10^5	74.7	102.3	124.8	227.1	56.9	308.8	33.0	10944.0	**92.8**
SALAD †	DINOv2-B	8448	28.6	6.9	**0**	6.9	29.6	88.0	0.6	1762	90.0
BoQ †	DINOv2-B	12288	12.4	7.6	**0**	7.6	28.3	100	0.9	2516	91.6
SALAD	**DINOv2-S**	8448	14.1	**2.4**	**0**	**2.4**	11.9	**22.9**	0.6	**1474**	87.9
BoQ	**DINOv2-S**	12288	7.2	4.2	**0**	4.2	15.9	34.4	0.9	2112	89.3
FoL	**DINOv2-S**	5×10^5	43.1	12.5	93.5	106.0	17.0	24.1	33.0	11454	88.9
EfficientVPR	**DINOv2-S**	**3456**	**6.1**	3.1	**0**	3.1	**7.7**	29.7	**0.2**	2360	90.3

Table 2. **Comprehensive comparison with SOTA methods across model scales**. † denotes the original version of the method reported by the authors. "Train Time" denotes train time per epoch. Memory footprint (Memory) is calculated on the MSLS-val dataset, which includes 18871 database images. Latency is measured on Pitts250k-test, using the same CPU and GPU (RTX A800) and is averaged over 5 identical runs. "Avg. Acc." denotes the average R@1 across seven datasets.

duces feature dimensionality (only 28.1% of BoQ's), leading to a markedly lower memory footprint (3× lower than SALAD, 4.5× lower than BoQ, and 165× lower than FoL), thereby greatly easing storage requirements in large-scale deployment. In cross-scale comparisons, EfficientVPR surpasses the DINOv2-B-based SALAD by 0.3% in Avg. Acc. with less than 34% model size and 26% training parameters. Compared to the high-dimensional DINOv2-B-based BoQ and the much larger DINOv2-L-based FoL, our method achieves competitive accuracy (within 1.3% and 2.5% Avg. Acc. loss) while delivering 2.5× and 73.3× speedup as well as 3.4× and 10.4× model compression. **Comparisons against one-stage SOTA methods across different model scales with same dimensions can be found in the Supplementary Material**, where our EfficientVPR surpasses all one-stage SOTA methods even larger backbones.

4.4. Ablation Study

To verify each component's impact, we conducted ablation studies on Pitts250-test, AmsterTime, and SVOX. These datasets cover key VPR challenges—viewpoint shifts, domain changes, and weather variations—enabling thorough evaluation. Since some versions replace the SGFS module with a linear layer, we resized all test images to 266 × 266 in the ablation experiments to ensure fair comparison.

4.4.1. Ablation on Scene-aware Visual Prompt Tuning

Comparison with other fine-tuning methods. We compared our SeneVPT with 9 distinct fine-tuning strategies, including completely frozen pretrained models, full-tuning, adapter-based method, LoRA, VPT-shallow, VPT-deep and two other efficient prompt-based finetuning methods (LoR-VP[27] and IAPT[56]). LoR-VP and IAPT are recent

prompt tuning methods: LoR-VP uses non-prefix prompts while IAPT employs encoder-based instance-aware prefix prompts (details in Supplementary Material).

In Tab.3, under frozen backbone settings, the model fails to adapt effectively to VPR, resulting in consistently poor performance across all datasets. While full fine-tuning improves accuracy on Pitts250k and SVOX-overcast,

Version	#Params. (M)	Pitts250k -test		Amster Time		SVOX Overcast	
		R@1	R@5	R@1	R@5	R@1	R@5
Frozen	0	89.8	96.3	43.1	64.8	89.1	96.8
Full-tuning	21.7	93.7	98.0	36.8	56.9	95.3	98.4
Adapter [22]	1.8	94.1	**98.6**	47.7	69.4	96.4	98.5
LoRA [23]	0.06	94.0	98.1	47.0	67.4	95.5	98.5
IAPT [56]	1.38	92.7	97.5	46.8	67.9	94.2	98.2
LoR-VP [27]	0.01	89.6	96.2	42.2	65.0	92.0	96.8
VPT-s [26]	0.01	93.0	97.5	47.7	68.7	94.7	98.3
VPT-d [26]	0.15	93.9	98.2	48.2	68.2	95.8	98.1
SceneVPT	0.15	**94.5**	98.2	**48.4**	**69.5**	**96.7**	**98.7**

Table 3. **Ablation on SceneVPT**. Given that SGFS relies on prompts peocessed by the backbone, it is replaced with a linear layer in all experiments of this set. All test images are resized to 266×266.

Version	Pitts250k -test		Amster Time		SVOX Overcast	
	R@1	R@5	R@1	R@5	R@1	R@5
Random	94.4	98.2	52.1	74.1	97.1	99.0
top-k	94.7	98.3	54.5	75.8	97.8	99.1
adaptive selection	**95.0**	**98.4**	**55.2**	**77.0**	**97.9**	**99.2**

Table 4. **Ablation on token selection strategy**. Random: randomly select prompts. Top-k: preserve only the top-k prompts with the highest attention scores (k=4, optimized through multiple trials). In contrast, our approach dynamically weights all tokens based on their attention values, effectively preserving more comprehensive information. Results demonstrate the superiority of our adaptive method across all datasets. All test images are resized to 266×266.

Version	Pitts250k -test		Amster Time		SVOX Overcast	
	R@1	R@5	R@1	R@5	R@1	R@5
MS-fc	94.2	98.1	47.1	69.2	96.1	98.6
MS+OLE-fc	94.4	98.2	48.0	69.2	96.3	98.7
MsIA+OLE-fc	94.5	98.2	48.4	69.5	96.7	99.0
MsIA+OLE-SGFS	**95.0**	**98.4**	**55.2**	**77.0**	**97.9**	**99.2**

Table 5. **Ablation on instance-dependent key local feature enhancement module**. We analyzed the role of each sub-module in the instance-dependent key local feature enhancement module by removing or replacing key components. All test images are resized to 266×266.

Version	Pitts250k -test		Amster Time		SVOX Overcast		Eynsham	
	R@1	R@5	R@1	R@5	R@1	R@5	R@1	R@5
with BoQ block	94.6	98.2	51.1	72.3	96.3	99.0	91.2	94.9
with SGFS	**95.0**	**98.4**	**55.2**	**77.0**	**97.9**	**99.2**	**91.6**	**95.1**

Table 6. **Ablation on SGFS.** "with BoQ block" refers to the variant where the SGFS module is replaced with a BoQ block, while "with SGFS" means our full model that retains SGFS. All test images are resized to 266×266.

Size	Pitts250k -test		Amster Time		SVOX Overcast	
	R@1	R@5	R@1	R@5	R@1	R@5
224×224	94.4	98.3	52.2	74.6	96.7	98.9
266×266	95.0	98.4	55.2	77.0	97.9	99.2
322×322	95.4	98.8	60.2	78.4	98.4	**99.4**
504×504	**95.8**	**98.7**	**61.4**	**82.0**	**98.5**	**99.4**

Table 7. **Ablation on inference image size**. While keeping the training image size fixed at 266×266, we test with images resized to varying sizes (in pixel).

it causes significant degradation on AmsterTime due to domain shift—revealing severe catastrophic forgetting of pretrained knowledge and compromised generalization capability. In contrast to other fine-tuning methods, our SceneVPT achieves the best R@1 on all datasets at a fraction of the parameters (just 8% of an Adapter). The accuracy gain on AmsterTime shows that SceneVPT better preserves pretrained features by generating task- and sample-specific prompts to guide adaptation, rather than directly modifying feature tokens like adapter-based methods. Moreover, compared with VPT-shallow, VPT-deep achieves better R@1 over Pitts250k-test and AmsterTime. It indicates that adjusting all layers of the model can lead to better performance. While, our SceneVPT outperforms VPT-deep by 0.6%, 0.2%, and 0.9% absolute R@1 gains on each dataset. Fig. 6 provides the further rationale for the results.

Ablation on the prompt selection strategy. To evaluate the contribution of adaptive prompt selection, we compare our Self-Adaptive Prompt Selection Mechanism with alternatives such as random selection and top-k. Results in Tab. 4 shows that compared with other two selection strategy, our dynamic weighted fusion approach proves more effective.

4.4.2. Ablation on Instance-dependent Key Local Feature Enhancement Module

Contribution of each component. In Tab. 5 evaluate the contribution of each sub-module in the instance-dependent key local feature enhancement module. Compared to the MS-fc baseline, incorporating the OLE module yields absolute improvements of 0.9% in R@1 on AmsterTime (significant domain and viewpoint variations), aligning with

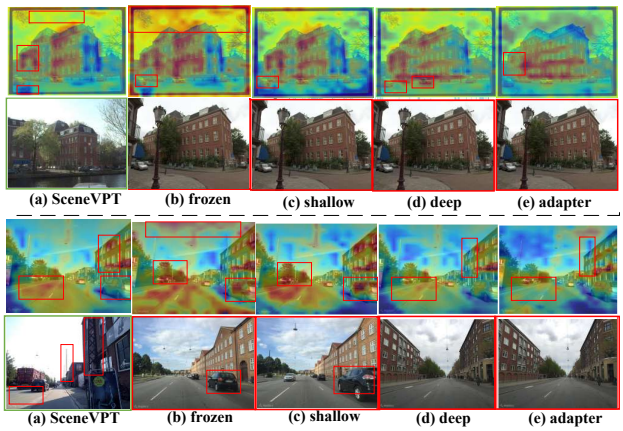


Figure 6. **Visualization of fine-tuning methods**. The first and third row visualizes features extracted by backbones fine-tuned with different methods. The second and fourth row shows their retrieval results (green boxes: correct matches; red boxes: errors). Frozen backbones attend to excessive task-irrelevant information. VPT-shallow improves but still has this issue. In the first group, street features are non-discriminative, whereas in the second group, they become discriminative features. Our SceneVPT enables dynamic adjustment of attention regions based on input images, while both VPT-deep and adapter-based methods fail to achieve this adaptability effectively.

OLE's design goal of enhancing feature robustness against domain and perspective changes. Adding the MsIA module to the MS-OLE-fc version further improves R@1 by 0.1%, 0.4%, and 0.4% on Pitts250k-test, AmsterTime, and SVOX-overcast, respectively. While the gain on Pitts250k-test is marginal, the consistent improvements on the other two datasets suggest that modeling cross-scale interactions helps address challenges like domain and environmental changes. The substitution of conventional linear projection with our SGFS module marked improvements on all evaluation datasets, especially over AmsterTime, where the model achieved an additional 6.8% boost in R@1.

Ablation on SGFS. Considering the structural similarity between our SGFS and the BoQ block, we substitute our SGFS with a BoQ block for comparison with our original version to verify the impact of the "sample-adaptive query" mechanism on the experimental results. We include the Eynsham dataset in this part to better assess scene variation. In Tab.6, the model with SGFS consistently outperforms the variant with BoQ block across all datasets, evidenced by its absolute R@1 improvements of 0.4% on Pitts250k-test, 4.1% on AmsterTime, and 1.5% on SVOX-overcast. These results demonstrate that the "sample-adaptive query" mechanism introduced by SGFS further enhance the discriminative power of the features, making the model more robust to viewpoint and domain changes.

4.4.3. Detailed Timing and Parameter Breakdown

We analyze each sub-module's parameter counts and timing breakdowns (Fig.7). Despite using DINOv2-S, the backbone still dominates both parameters and latency, highlighting that adopting a more efficient backbone is key to further improving VPR efficiency. Although prompt-tuning adds some latency (0.31ms/image), it is negligible (adding only about 2.5s for all queries in Pitts250k). Among the aggregation sub-modules (MsIA, OLE, and SGFS), their parameter and latency contributions are relatively small compared to the backbone. Even the largest sub-module, MsIA, has only about 0.2 of the backbone's parameters and an extremely low latency of about 0.1ms, demonstrating the high efficiency of our sub-module design.

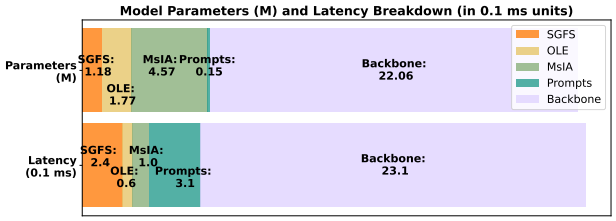


Figure 7. **Parameters and detailed timing breakdown by sub-module**. Results are based on the average of 5 identical runs, measured on Pitts250k using the same CPU and GPU (RTX A800). Latency (0.1ms) and Parameters (Millions).

4.4.4. Ablation on Image Size

Inference image size. Notably, as some methods use higher image resolutions, we evaluate our model across varying input sizes to isolate the impact of this factor. In Tab.7, beyond 322×322 performance gains plateau, we select this size to maintain a trade-off between accuracy and efficiency.

Training image size. In Tab.7, we evaluate how training image resolution affects our model's performance. The 266×266 resolution performs best overall, while larger sizes only improve results on certain datasets.

Size	Pitts250k -test		Amster Time		SVOX Overcast	
	R@1	R@5	R@1	R@5	R@1	R@5
224×224	94.8	98.5	57.8	78.8	97.7	98.9
266×266	**95.4**	**98.8**	60.2	78.4	**98.4**	**99.4**
280×280	95.2	98.6	**61.4**	**82.0**	98.2	99.2

Table 8. **Ablation on training image size**. While training with varying sizes, we test with images fixed at 322×322 (in pixel).

5. Conclusions

In this paper, we propose an efficient one-stage method for VPR. By developing scene-aware fine-tuning method and an instance-dependent key local feature enhancement module, our model enhances discriminative power while drastically reducing model complexity. Experiments over 7 mainstream benchmarks show that our method sets a new SOTA with DINOv2-S and delivers an efficient model with compact dimensionality that remains competitive with the large-model SOTA.

6. Acknowledgments

This work is supported by the National Natural Science Foundation of China under Grant Number 62406312 and 62476264, the Beijing Natural Science Foundation under Grant Number 4244098, and the Science Foundation of the Chinese Academy of Sciences.

References

[1] Armen Aghajanyan, Luke Zettlemoyer, and Sonal Gupta. Intrinsic dimensionality explains the effectiveness of language model fine-tuning, 2020. 3

[2] Amar Ali-bey, Brahim Chaib-draa, and Philippe Giguère. MixVPR: feature mixing for visual place recognition. In *WACV*, pages 2998–3007, 2023. 1, 2, 6

[3] Amar Ali-bey, Brahim Chaib-draa, and Philippe Giguère. Global proxy-based hard mining for visual place recognition, 2023. 2

[4] Amar Ali-Bey, Brahim Chaib-draa, and Philippe Giguere. Boq: A place is worth a bag of learnable queries. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17794–17803, 2024. 1, 2, 4, 6, 7

[5] Amar Ali-bey et al. Gsv-cities: Toward appropriate supervised visual place recognition. *Neurocomputing*, 513:194–203, 2022. 5, 4

[6] Zhulin An, Xinqiang Yu, Chu Wang, Yinlong Zhang, and Chunhe Song. Embodied intelligence: Recent advances and future perspectives. *The Innovation Informatics*, 1(1): 100008, 2025. 1

[7] Relja Arandjelovic, Petr Gronat, Akihiko Torii, Tomas Pajdla, and Josef Sivic. Netvlad: Cnn architecture for weakly supervised place recognition. In *CVPR*, pages 5297–5307, 2016. 6, 5

[8] H. Bay, A. Ess, T. Tuytelaars, and L. V. Gool. Speeded-up robust features (surf). *Computer vision and image understanding*, 110(3):346–359, 2008. 2

[9] Gabriele Berton, Carlo Masone, and Barbara Caputo. Rethinking visual geo-localization for large-scale applications. In *CVPR*, pages 4878–4888, 2022. 1, 6

[10] Gabriele Berton, Gabriele Trivigno, Barbara Caputo, and Carlo Masone. Eigenplaces: Training viewpoint robust models for visual place recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11080–11090, 2023. 6

[11] Gabriele Moreno Berton, Valerio Paolicelli, Carlo Masone, and Barbara Caputo. Adaptive-attentive geolocalization from few queries: A hybrid approach. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 2918–2927, 2021. 5

[12] Bingyi Cao, Andre Araujo, and Jack Sim. Unifying deep local and global features for image search. In *European conference on computer vision*, pages 726–743. Springer, 2020. 2

[13] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. In *European conference on computer vision*, pages 213–229. Springer, 2020. 7

[14] Mark Joseph Cummins and Paul Newman. Highly scalable appearance-only slam - fab-map 2.0. In *Robotics: Science and Systems*, 2009. 5

[15] Anurag Das, Xinting Hu, Li Jiang, and Bernt Schiele. Mtaclip: Language-guided semantic segmentation with mask-text alignment. In *European Conference on Computer Vision*, pages 39–56. Springer, 2024. 2

[16] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2020. 3

[17] Martin A. Fischler and Robert C. Bolles. Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM*, 24(6):381–395, 1981. 1

[18] Yixiao Ge, Haibo Wang, Feng Zhu, Rui Zhao, and Hongsheng Li. Self-supervising fine-grained region similarities for large-scale image localization. In *ECCV*, pages 369–386. Springer, 2020. 6

[19] Zeyu Han, Chao Gao, Jinyang Liu, Jeff Zhang, and Sai Qian Zhang. Parameter-efficient fine-tuning for large models: A comprehensive survey. *arXiv preprint arXiv:2403.14608*, 2024. 2

[20] Stephen Hausler, Sourav Garg, Ming Xu, Michael Milford, and Tobias Fischer. Patch-netvlad: Multi-scale fusion of locally-global descriptors for place recognition. In *CVPR*, pages 14141–14152, 2021. 2

[21] Junxian He, Chunting Zhou, Xuezhe Ma, Taylor Berg-Kirkpatrick, and Graham Neubig. Towards a unified view of parameter-efficient transfer learning. *arXiv preprint arXiv:2110.04366*, 2021. 2

[22] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-efficient transfer learning for nlp. In *International conference on machine learning*, pages 2790–2799. PMLR, 2019. 2, 7

[23] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021. 7

[24] Libo Huang, Zhulin An, Chuanguang Yang, Boyu Diao, Fei Wang, Yan Zeng, Zhifeng Hao, and Yongjun Xu. Preprompt: Predictive prompting for class incremental learning, 2025. 2

[25] Sergio Izquierdo and Javier Civera. Optimal transport aggregation for visual place recognition. In *Proceedings of the ieee/cvf conference on computer vision and pattern recognition*, pages 17658–17668, 2024. 1, 2, 6

[26] Menglin Jia, Luming Tang, Bor-Chun Chen, Claire Cardie, Serge Belongie, Bharath Hariharan, and Ser-Nam Lim. Visual prompt tuning. In *European conference on computer vision*, pages 709–727. Springer, 2022. 2, 7

[27] Can Jin, Ying Li, Mingyu Zhao, Shiyu Zhao, Zhenting Wang, Xiaoxiao He, Ligong Han, Tong Che, and Dimitris N. Metaxas. Lor-vp: Low-rank visual prompting for efficient vision model adaptation, 2025. 6, 7

[28] Hervé Jégou, Matthijs Douze, Cordelia Schmid, and Patrick Pérez. Aggregating local descriptors into a compact image representation. In *CVPR*, pages 3304–3311, 2010. 2

[29] Nikhil Keetha, Avneesh Mishra, Jay Karhade, Krishna Murthy Jatavallabhula, Sebastian Scherer, Madhava Krishna, and Sourav Garg. Anyloc: Towards universal visual place recognition. *IEEE Robotics and Automation Letters*, 9 (2):1286–1293, 2023. 1, 2

[30] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E. Hinton. Imagenet classification with deep convolutional neural networks. *Commun. ACM*, 60(6):84–90, 2017. 1, 2

[31] Brian Lester, Rami Al-Rfou, and Noah Constant. The power of scale for parameter-efficient prompt tuning. *arXiv preprint arXiv:2104.08691*, 2021. 2

[32] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, et al. Grounding dino: Marrying dino with grounded pre-training for open-set object detection. In *European conference on computer vision*, pages 38–55. Springer, 2024. 2

[33] Feng Lu, Xiangyuan Lan, Lijun Zhang, Dongmei Jiang, Yaowei Wang, and Chun Yuan. Cricavpr: Cross-image correlation-aware representation learning for visual place recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16772–16782, 2024. 1, 2, 6

[34] Feng Lu, Lijun Zhang, Xiangyuan Lan, Shuting Dong, Yaowei Wang, and Chun Yuan. Towards seamless adaptation of pre-trained models for visual place recognition. In *ICLR*, 2024. 2, 6, 7

[35] Carlo Masone and Barbara Caputo. A survey on deep visual place recognition. *IEEE Access*, 9:19516–19547, 2021. 1

[36] Eva Mohedano, Kevin McGuinness, Noel E O'Connor, Amaia Salvador, Ferran Marques, and Xavier Giró-i Nieto. Bags of local convolutional features for scalable instance search. In *Proceedings of the 2016 ACM on international conference on multimedia retrieval*, pages 327–331, 2016. 2

[37] Hyeonwoo Noh, Andre Araujo, Jack Sim, Tobias Weyand, and Bohyung Han. Large-scale image retrieval with attentive deep local features. In *Proceedings of the IEEE international conference on computer vision*, pages 3456–3465, 2017. 1

[38] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023. 2

[39] Guohao Peng, Jun Zhang, Heshan Li, and Danwei Wang. Attentional pyramid pooling of salient visual residuals for place recognition. In *ICCV*, pages 885–894, 2021. 1

[40] Filip Radenović, Giorgos Tolias, and Ondřej Chum. Fine-tuning cnn image retrieval with no human annotation. *IEEE transactions on pattern analysis and machine intelligence*, 41(7):1655–1668, 2018. 4

[41] Akihiko Torii, Josef Sivic, Tomas Pajdla, and Masatoshi Okutomi. Visual place recognition with repetitive structures. In *CVPR*, pages 883–890, 2013. 5, 4

[42] Changwei Wang, Shunpeng Chen, Yukun Song, Rongtao Xu, Zherui Zhang, Jiguang Zhang, Haoran Yang, Yu Zhang, Kexue Fu, Shide Du, et al. Focus on local: Finding reliable discriminative regions for visual place recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 7536–7544, 2025. 6, 7

[43] Heng Wang and Cordelia Schmid. Action recognition with improved trajectories. In *Proceedings of the IEEE international conference on computer vision*, pages 3551–3558, 2013. 2

[44] Xun Wang, Xintong Han, Weilin Huang, Dengke Dong, and Matthew R Scott. Multi-similarity loss with general pair weighting for deep metric learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5022–5030, 2019. 5

[45] Yuwei Wang, Yuanying Qiu, Peitao Cheng, and Junyu Zhang. Transformer-based descriptors with fine-grained region supervisions for visual place recognition. *Knowledge-Based Systems*, 280:110993, 2023. 1

[46] Yuzhu Wang, Lechao Cheng, Chaowei Fang, Dingwen Zhang, Manni Duan, and Meng Wang. Revisiting the power of prompt for visual tuning. *arXiv preprint arXiv:2402.02382*, 2024. 2

[47] Frederik Warburg, Soren Hauberg, Manuel Lopez-Antequera, Pau Gargallo, Yubin Kuang, and Javier Civera. Mapillary street-level sequences: A dataset for lifelong place recognition. In *CVPR*, pages 2626–2635, 2020. 5, 4

[48] Zhuofeng Wu, Sinong Wang, Jiatao Gu, Rui Hou, Yuxiao Dong, V. G. Vinod Vydiswaran, and Hao Ma. Idpg: An instance-dependent prompt generation method, 2022. 2

[49] Xi Xiao, Yunbei Zhang, Xingjian Li, Tianyang Wang, Xiao Wang, Yuxiang Wei, Jihun Hamm, and Min Xu. Visual instance-aware prompt tuning, 2025. 2

[50] Chuanguang Yang, Helong Zhou, Zhulin An, Xue Jiang, Yongjun Xu, and Qian Zhang. Cross-image relational knowledge distillation for semantic segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 12319–12328, 2022. 2

[51] Chuanguang Yang, Zhulin An, Libo Huang, Junyu Bi, Xinqiang Yu, Han Yang, Boyu Diao, and Yongjun Xu. Clip-kd: An empirical study of clip model distillation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15952–15962, 2024. 2

[52] Burak Yildiz, Seyran Khademi, Ronald Maria Siebes, and Jan van Gemert. Amstertime: A visual place recognition benchmark dataset for severe domain shift. *arXiv preprint arXiv:2203.16291*, 2022. 5

[53] Hao Zhang, Xin Chen, Heming Jing, Yingbin Zheng, Yuan Wu, and Cheng Jin. Etr: An efficient transformer for reranking in visual place recognition. In *WACV*, pages 5665–5674, 2023. 1

[54] Xiwu Zhang, Lei Wang, and Yan Su. Visual place recognition: A survey from deep learning perspective. *Pattern Recognition*, 113:107760, 2021. 1

[55] Sijie Zhu, Linjie Yang, Chen Chen, Mubarak Shah, Xiaohui Shen, and Heng Wang. R^2Former: unified retrieval and reranking transformer for place recognition. In *CVPR*, pages 19370–19380, 2023. 6, 7

[56] Wei Zhu, Aaron Tian, Congrui Yin, Yuan Ni, Xiaoling Wang, and Guotong Xie. IAPT: Instance-aware prompt tuning for

large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14285–14304, Bangkok, Thailand, 2024. Association for Computational Linguistics. 2, 6, 7